

At level 1, the Kashiwara-crystal string of $v_{k',e} = P_e^{k'}|\emptyset\rangle$ has length at most one:

$$\varepsilon_i(v_{k',e}) \leq 1 \text{ for all } i \in \mathbb{Z}/e\mathbb{Z}$$

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Abstract

Let $e \geq 2$ and let $\mathcal{F}_e$ denote the level-1 Uglov q -Fock space of $U_q(\widehat{\mathfrak{sl}}_e)$, with standard basis $\{|\lambda\rangle\}$ indexed by partitions. Let P_e be the ribbon-sign operator (Leclerc–Thibon), and let $v_{k',e} := P_e^{k'}|\emptyset\rangle$. Let $(\mathcal{L}, \{[\mu]\})$ denote the Kashiwara crystal lattice of $\mathcal{F}_e$, and let $\{\tilde{e}_i, \tilde{f}_i\}_{i \in \mathbb{Z}/e\mathbb{Z}}$ be the Kashiwara operators.

We prove:

(C5, $\ell = 1$) *Kashiwara string-length bound.* For every $e \geq 2$, every $k' \geq 1$, and every $i \in \mathbb{Z}/e\mathbb{Z}$,

$$\varepsilon_i(v_{k',e}) = \max\{n \geq 0 : \tilde{e}_i^n(v_{k',e} \bmod q\mathcal{L}) \neq 0\} \leq 1.$$

The proof is elementary and Fock-native: it factors through a closed expansion of $[v_{k',e}]$ in the crystal basis at $q = 0$, together with a purely combinatorial signature-word analysis for partitions all of whose parts are divisible by e . Gerber’s bicrystal machinery [3] (which is stated only for $\ell \geq 2$) is not needed at level 1; the role played by his Cor. 5.3 in shaping the intuition is discussed in Remark 12. As a corollary we obtain that $v_{k',e} \bmod q\mathcal{L}$ is a $\mathbb{Z}$ -linear combination of crystal-basis elements $[e\lambda]$ indexed by partitions $\lambda \vdash k'$, with coefficients equal to the number f^λ of standard Young tableaux of shape λ (Theorem 4).

We verify $\varepsilon_i(v_{k',e}) \leq 1$ empirically on 14 target triples plus 15 extended triples (e, k', i) , verify the expansion $[v_{k',e}] = \sum_\lambda f^\lambda [e\lambda]$ on 17 pairs (e, k') (all with $ek' \leq 16$), and verify the individual bound $\varepsilon_i(e\lambda) \leq 1$ on 102 partitions λ across ranges $e \in \{2, 3, 4, 5\}$, $|\lambda| \leq 7$ — all pass.

1 Setup

1.1 Level-1 Uglov Fock space

Fix $e \geq 2$. Let $\mathcal{F}_e$ denote the level-1 Uglov q -Fock space of $U_q(\widehat{\mathfrak{sl}}_e)$, viewed as a free $\mathbb{Z}[q, q^{-1}]$ -module with standard basis $\{|\lambda\rangle : \lambda \in \Pi\}$ indexed by all partitions $\lambda = (\lambda_1 \geq \lambda_2 \geq \dots \geq 0)$. The Chevalley generators $e_i, f_i, K_i^{\pm 1}$ act on the standard basis by Leclerc–Thibon’s formulas (see [1, 2]); we recall the formulas for e_i and f_i that we need:

$$\begin{aligned} f_i|\lambda\rangle &= \sum_{A \in \mathbf{A}_i(\lambda)} q^{N_i^r(\lambda, A)} |\lambda + A\rangle, \\ e_i|\lambda\rangle &= \sum_{R \in \mathbf{R}_i(\lambda)} q^{-N_i^\ell(\lambda, R)} |\lambda - R\rangle, \end{aligned}$$

where $\mathbf{A}_i(\lambda)$ (resp. $\mathbf{R}_i(\lambda)$) is the set of addable (resp. removable) i -nodes of λ , and N_i^r, N_i^ℓ are the usual signature-based exponents. Residues of boxes are computed as $\text{res}(r, c) = c - r \bmod e$ using 0-indexed rows and columns.

1.2 The ribbon operator P_e and the vector $v_{k',e}$

Define the *ribbon-sign operator* P_e on $\mathcal{F}_e$ by

$$P_e|\lambda\rangle = \sum_{\mu=\lambda+e\text{-ribbon}} (-q)^{h(\mu/\lambda)}|\mu\rangle,$$

where the sum runs over all μ obtained from λ by adding a connected ribbon of size e , and $h(\mu/\lambda) = (\text{rows spanned}) - 1$ is the ribbon height. Set

$$v_{k',e} := P_e^{k'}|\emptyset\rangle \in \mathcal{F}_e.$$

1.3 Kashiwara crystal lattice and operators

Let $A := \mathbb{Z}_{(q)}$ denote the local ring at $q = 0$ of $\mathbb{Z}[q]$ (polynomials $g(q) \in \mathbb{Z}[q]$ localized by inverting those with $g(0) \neq 0$).

Definition 1. The *Kashiwara crystal lattice* of $\mathcal{F}_e$ is

$$\mathcal{L} := \bigoplus_{\lambda \in \Pi} A \cdot |\lambda\rangle \subset \mathcal{F}_e \otimes_{\mathbb{Z}[q,q^{-1}]} A[q^{-1}].$$

The reduction $\mathcal{L}/q\mathcal{L}$ is a free $\mathbb{Z}$ -module with basis $\{[|\lambda\rangle] : \lambda \in \Pi\}$, called the crystal basis. The Kashiwara operators $\tilde{e}_i, \tilde{f}_i$ act on $\mathcal{L}/q\mathcal{L}$ by the good-node rule (Kleshchev–Uglov): for $\mu \in \Pi$,

$$\tilde{e}_i[\mu] = \begin{cases} [\mu \setminus \{n\}] & \text{if } \mu \text{ has a good removable } i\text{-node } n, \\ 0 & \text{otherwise,} \end{cases}$$

where a good removable i -node is determined by reading the i -signature of μ in bottom-up row order and cancelling adjacent RA pairs; the leftmost surviving R is the good one.¹ Extend $\tilde{e}_i, \tilde{f}_i$ to $\mathbb{Z}$ -linear operators on $\mathcal{L}/q\mathcal{L}$.

For $v \in \mathcal{L}$, define

$$\varepsilon_i(v) := \max\{n \geq 0 : \tilde{e}_i^n([v]) \neq 0\},$$

with $\varepsilon_i(0) := -\infty$.

That the good-node rule computes the Kashiwara operators on all of $\mathcal{L}/q\mathcal{L}$ (not merely the sub-crystal $V(\Lambda_0) \subset \mathcal{F}_e$) is Ariki–Uglov; see [1, 2]. In particular the operators are $\mathbb{Z}$ -linear, so

$$\tilde{e}_i\left(\sum_{\mu} c_{\mu}[\mu]\right) = \sum_{\mu} c_{\mu} \tilde{e}_i[\mu]. \quad (1)$$

2 Main theorem

Theorem 2 (C5 at level 1). *For every $e \geq 2$, every $k' \geq 1$, and every $i \in \mathbb{Z}/e\mathbb{Z}$,*

$$\varepsilon_i(v_{k',e}) \leq 1.$$

Concretely: $\tilde{e}_i^2([v_{k',e}]) = 0$ in $\mathcal{L}/q\mathcal{L}$.

The proof occupies Sections 3–6.

¹We use A (Addable) and R (Removable) for signature symbols, matching the code in `probe4_canonical.py`.

3 $v_{k',e}$ lies in the Kashiwara lattice

Proposition 3. $v_{k',e} \in \mathcal{L}$ for every $e \geq 2$ and $k' \geq 0$.

Proof. The Uglov ribbon formula $P_e|\lambda\rangle = \sum_{\mu} (-q)^{h(\mu/\lambda)} |\mu\rangle$ has coefficients $(-q)^h$ with $h \geq 0$, so all coefficients lie in $\mathbb{Z}[q] \subset A$. Since $|\emptyset\rangle \in \mathcal{L}$ trivially and $\mathcal{L}$ is closed under P_e (its matrix entries lie in A), $v_{k',e}$ is obtained from $|\emptyset\rangle$ by k' successive applications of P_e and stays in $\mathcal{L}$. $\square$

4 Closed form for $[v_{k',e}]$ modulo $q\mathcal{L}$

For a partition $\lambda = (\lambda_1, \dots, \lambda_\ell)$ let $e\lambda := (e\lambda_1, e\lambda_2, \dots, e\lambda_\ell)$. Let f^λ denote the number of standard Young tableaux of shape λ (equivalently, the number of chains $\emptyset = \nu^{(0)} \subset \nu^{(1)} \subset \dots \subset \nu^{(k')} = \lambda$ where each $\nu^{(t+1)}$ is obtained from $\nu^{(t)}$ by adding one box).

Theorem 4 (Expansion of $v_{k',e}$ in the crystal basis at $q = 0$). *For every $e \geq 2$ and $k' \geq 0$,*

$$[v_{k',e}] = \sum_{\lambda \vdash k'} f^\lambda [e\lambda] \quad \text{in } \mathcal{L}/q\mathcal{L}.$$

In particular, $\text{supp}([v_{k',e}]) \subseteq \{e\lambda : \lambda \vdash k'\}$ inside the crystal basis.

Proof. Write $P_e^{(0)}$ for the specialization $P_e|_{q=0}$, acting on the $\mathbb{Z}$ -module $\mathcal{L}/q\mathcal{L}$. Since the Uglov formula weights each added e -ribbon by $(-q)^h$, only the height-0 ribbons survive at $q = 0$:

$$P_e^{(0)}[|\lambda\rangle] = \sum_{\substack{\mu = \lambda + e\text{-ribbon} \\ h(\mu/\lambda) = 0}} [|\mu\rangle].$$

A height-0 e -ribbon is a horizontal e -strip contained in a single row.

Restriction to partitions divisible by e . We show by induction on k' that the support of $P_e^{(0)^{k'}}[|\emptyset\rangle]$ lies in $\{[e\lambda] : \lambda \vdash k'\}$. Base $k' = 0$: support is $\{[|\emptyset\rangle]\} = \{[e \cdot \emptyset]\}$. Inductive step: assume the support after k' steps consists of $\mu = e\nu$ for various $\nu \vdash k'$. A height-0 e -ribbon added to $\mu = e\nu$ must be a single row of e boxes; it can be added (a) by extending an existing row μ_r from $e\nu_r$ to $e(\nu_r + 1)$, valid iff $r = 0$ or $\nu_{r-1} > \nu_r$; or (b) as a new row ($r = \ell(\nu)$, length e). In case (a), the resulting partition has all parts divisible by e and equals $e\nu'$ where ν' is ν with ν_r incremented by 1. In case (b), the result equals $e\nu'$ where $\nu' = (\nu_1, \dots, \nu_{\ell(\nu)}, 1)$. Either way the support after $k' + 1$ steps lies in $\{e\nu' : \nu' \vdash k' + 1\}$.

Coefficient of $[e\lambda]$. The multiplicity of $[e\lambda]$ in $P_e^{(0)^{k'}}[|\emptyset\rangle]$ equals the number of sequences $\nu^{(0)} = \emptyset, \nu^{(1)}, \dots, \nu^{(k')} = \lambda$ of partitions with $\nu^{(t+1)}$ obtained from $\nu^{(t)}$ by (i) extending some existing row by one box in an admissible way (row $r = 0$ or $\nu_{r-1}^{(t)} > \nu_r^{(t)}$), or (ii) appending a new row of length one. But these are precisely the possible growth steps by an addable node of $\nu^{(t)}$; the sequence is a standard Young tableau of shape λ read in growth order. Hence the multiplicity is f^λ . $\square$

Example 5. The small expansions:

$$\begin{aligned} [v_{2,2}] &= [|(4)\rangle] + [|(2,2)\rangle] \\ [v_{3,2}] &= [|(6)\rangle] + 2[|(4,2)\rangle] + [|(2,2,2)\rangle] \\ [v_{2,3}] &= [|(6)\rangle] + [|(3,3)\rangle] \\ [v_{3,3}] &= [|(9)\rangle] + 2[|(6,3)\rangle] + [|(3,3,3)\rangle] \\ [v_{2,4}] &= [|(8)\rangle] + [|(4,4)\rangle] \end{aligned}$$

Coefficients are exactly $f^{(2)} = f^{(1,1)} = 1$, $f^{(3)} = f^{(1,1,1)} = 1$, $f^{(2,1)} = 2$, matching Theorem 4. All five are computed in `probes/2026-08-11-C5-gerber/epsilon_i_probe.log`.

5 The individual bound: $\varepsilon_i([e\lambda]) \leq 1$

Lemma 6 (Validity dictionary). *Let $\mu = e\lambda$, $L = \ell(\lambda)$, with 0-indexed rows and the conventions $\mu_{-1} = +\infty$, $\mu_L = 0$. For $0 \leq r \leq L$ put*

$$S_r := [\mu_{r-1} > \mu_r] = [\lambda_{r-1} > \lambda_r].$$

Then:

1. μ has an addable node in row $r \iff S_r$;
2. μ has a removable node in row $r-1 \iff S_r$ (for $1 \leq r \leq L$);
3. when they exist, both nodes have residue $-r \bmod e$.

In particular S_0 and S_L are both true, so case (a) below supplies an unpaired $\mathbf{A}$ and case (c) a genuine pair.

Proof. (1) By definition an addable node in row r exists exactly when row r is a strict step, i.e. $r = 0$ (where $\mu_{-1} = +\infty > \mu_0$) or $\mu_{r-1} > \mu_r$; for $r = L$ this is $\mu_{L-1} > 0 = \mu_L$, true. All three cases are the single statement S_r .

(2) By definition a removable node in row $r-1$ exists exactly when $\mu_{r-1} > \mu_r$ — for $r-1 = L-1$ this reads $\mu_{L-1} > 0 = \mu_L$, again true. *This is literally the same inequality between the same two consecutive parts as in (1).* The apparent index shift is illusory: removability looks one row *down*, addability looks one row *up*, and shifting the removable row from r to $r-1$ cancels that difference exactly. No property of $e\lambda$ is used; the coupling holds for any partition.

Since $\mu_{r-1} = e\lambda_{r-1}$ and $\mu_r = e\lambda_r$ and $e > 0$, the inequality $\mu_{r-1} > \mu_r$ is equivalent to $\lambda_{r-1} > \lambda_r$.

(3) *Here* divisibility by e is used, and it is what makes the two residues agree. The addable node of row r sits at column $\mu_r \equiv 0 \pmod{e}$, so its residue is $\mu_r - r \equiv -r$. The removable node of row $r-1$ sits at column $\mu_{r-1} - 1 \equiv -1 \pmod{e}$, so its residue is $(\mu_{r-1} - 1) - (r-1) \equiv -r$. Both are $\equiv -r$, hence both equal i precisely when $r \equiv -i \pmod{e}$. $\square$

Remark 7. *Lemma 6(3) is the only place where $\mu = e\lambda$ (as opposed to an arbitrary partition) is used, and it is used only to align the two residues. The coupling $(1) \Leftrightarrow (2)$ is completely general. This is the precise sense in which “inflate by e ” is what makes the signature pair up: it does not create the pairing, it makes the two members of each pre-existing pair carry the same residue.*

Proposition 8 (Individual crystal string bound). *Let λ be any partition and $i \in \mathbb{Z}/e\mathbb{Z}$. Then $\varepsilon_i([|e\lambda|]) \leq 1$; in fact $\varepsilon_i([|e\lambda|]) \in \{0, 1\}$.*

Proof. Set $\mu := e\lambda$, $L := \ell(\lambda)$. We analyze the addable and removable i -nodes of μ and reduce the resulting i -signature directly.

Step 1: Row-residue classification. With 0-indexed rows and columns, an addable node in row r of μ (existing when the row is a strict step, i.e., $r = 0$ or $\mu_{r-1} > \mu_r$) sits at column μ_r with residue $-r \bmod e$; a removable node in row r (existing when $r = L-1$ or $\mu_{r+1} < \mu_r$) sits at column $\mu_r - 1$ with residue $-r - 1 \bmod e$. In addition, if $r = L$ we may add a new row of length 1 with residue $-L \bmod e$. Since $e\lambda$ has all parts divisible by e , the strict-step condition $\mu_{r-1} > \mu_r$ is equivalent to $\lambda_{r-1} > \lambda_r$.

For fixed $i \in \mathbb{Z}/e\mathbb{Z}$:

- candidate addable- i -rows: r with $r \equiv -i \pmod{e}$ and $0 \leq r \leq L$;
- candidate removable- i -rows: r with $r \equiv -(i+1) \pmod{e}$ and $0 \leq r \leq L-1$.

Note that within any single row, addable and removable nodes carry different residues (assuming $e \geq 2$), so each row contributes at most one of A, R to the i -signature.

Step 1b: the validity coupling. Lemma 6 is the load-bearing fact of the whole argument: the addable node of row r and the removable node of row $r-1$ are governed by *the same* boolean. If they were governed by different booleans, lone R symbols would appear in the signature word and the bound $\varepsilon_i \leq 1$ would fail. (This step was asserted without proof in the first version of this note; the gap was identified by Lyra in her review of 30 August 2026 and is closed here.)

Step 2: Pairing structure. The rows $r' \equiv -i-1 \pmod{e}$ are exactly the rows $r' = r-1$ for r a candidate addable- i -row with $r \geq 1$. Thus every candidate addable- i -row $r \geq 1$ is immediately preceded (in top-down order) by a candidate removable- i -row $r-1$. By Lemma 6 the residues match in these adjacent rows, and both the addable-at- r and the removable-at- $(r-1)$ existence conditions are the single boolean $S_r = [\lambda_{r-1} > \lambda_r]$.

We split into cases, in each case reading Lemma 6 at the relevant r :

(a) $r = 0$, $i \equiv 0 \pmod{e}$. The addable at $(0, \mu_0)$ is always present (strict step at the top). No removable in “row -1 ” exists. This contributes a lone A symbol.

(b) $1 \leq r \leq L-1$, $r \equiv -i \pmod{e}$. By Lemma 6, addable-at- r and removable-at- $(r-1)$ are both present if S_r holds and both absent if it does not; there is no case in which exactly one of them occurs. The two nodes sit in rows r (deeper) and $r-1$ (shallower), so in bottom-up order this contributes AR — and otherwise nothing.

(c) $r = L$, $L \equiv -i \pmod{e}$. Here S_L is true unconditionally ($\mu_{L-1} > 0 = \mu_L$), so by Lemma 6 the addable at the new row $(L, 0)$ and the removable at row $L-1$ are both present. This contributes AR in bottom-up order.

Between two consecutive candidate rows (which are separated by exactly e in the row index), rows of intermediate index carry residues in $\mathbb{Z}/e\mathbb{Z} \setminus \{-i, -i-1\}$, contributing nothing to the i -signature.

Step 3: Bottom-up signature word. Reading the i -signature of μ in bottom-up row order (largest row index first, matching `order='bottom_up'` in `probe4_canonical.i_nodes`) gives a word of the form

$$\text{AR} \cdot \text{AR} \cdots \text{AR} \cdot \text{A}^?$$

where each “AR” is contributed by case (b) or (c), and the optional trailing “A[?]” is the lone A from case (a) (present iff $i \equiv 0 \pmod{e}$).

Step 4: Cancellation. Kashiwara’s cancellation rule for the bottom-up signature is: while reading tokens left-to-right, cancel any A that immediately follows an R on the stack. Concretely (in pseudocode):

```
for tok in seq:
    if tok == 'A' and stack and stack[-1] == 'R':
        stack.pop()      # cancel R-then-A
    else:
        stack.append(tok)
```

The surviving stack always has A’s at the bottom and R’s on top (no R can be immediately followed by an A that survives). $\varepsilon_i(\mu)$ equals the number of surviving R’s.

Case A: the signature has $n \geq 0$ blocks AR and *no* trailing A. After processing the first block, the stack is $[A, R]$; each subsequent AR block first cancels the top R (leaving $[A]$), then pushes R (leaving $[A, R]$). By induction the final stack is $[A, R]$ if $n \geq 1$, and empty if $n = 0$. Thus $\varepsilon_i(\mu) \in \{0, 1\}$.

Case B: the signature has $n \geq 0$ blocks AR followed by a trailing A. After processing the AR-blocks, the stack is $[A, R]$ (or empty if $n = 0$). Reading the trailing A: it cancels the top R, leaving $[A]$ (or if $n = 0$, pushing to $[A]$). No surviving R. Thus $\varepsilon_i(\mu) = 0$.

In both cases $\varepsilon_i(\mu) \leq 1$. $\square$

Remark 9. *The proof shows $\varepsilon_0([e\lambda]) = 0$ for every λ (since $r = 0$ always contributes a trailing lone A when $i = 0$). Empirically this matches all 17 test cases (see `individual_and_expansion_check.log`).*

Example 10. Take $e = 2$, $\lambda = (5, 3, 2, 1)$ so $\mu = e\lambda = (10, 6, 4, 2)$, $L = 4$. Compute ε_1 :

- candidate rows $r \equiv -1 \equiv 1 \pmod{2}$: $r \in \{1, 3\}$;
- $r = 1$: $\lambda_0 > \lambda_1$ ($5 \not\leq 3$) $\Rightarrow$ block $[A(1, 6), R(0, 9)]$;
- $r = 3$: $\lambda_2 > \lambda_3$ ($2 \not\leq 1$) $\Rightarrow$ block $[A(3, 2), R(2, 3)]$.

Bottom-up signature: $[A(3, 2), R(2, 3), A(1, 6), R(0, 9)] = ARAR$. Cancellation: push A; push R; A cancels top R; push R. Final stack: $[A, R]$. So $\varepsilon_1(\mu) = 1$, with good removable node $(2, 3)$.

6 Combining the pieces

Proof of Theorem 2. By Proposition 3, $v_{k',e} \in \mathcal{L}$, so the reduction $[v_{k',e}] \in \mathcal{L}/q\mathcal{L}$ is defined. By Theorem 4,

$$[v_{k',e}] = \sum_{\lambda \vdash k'} f^\lambda [|e\lambda\rangle].$$

By Proposition 8, $\tilde{e}_i^2[|e\lambda\rangle] = 0$ for every λ . By $\mathbb{Z}$ -linearity of $\tilde{e}_i^2$ (equation (1)),

$$\tilde{e}_i^2([v_{k',e}]) = \sum_{\lambda \vdash k'} f^\lambda \tilde{e}_i^2([|e\lambda\rangle]) = 0.$$

Hence $\varepsilon_i(v_{k',e}) \leq 1$. $\square$

Remark 11 (Sharpness). *The bound $\varepsilon_i(v_{k',e}) \leq 1$ is sharp for many (e, k', i) . For instance, at $(e, k', i) = (2, 2, 1)$ we have $\tilde{e}_1([v_{k',e}]) = [| (3) \rangle] \neq 0$, so $\varepsilon_1(v_{2,2}) = 1$. See the table in Section 7.*

Remark 12 (Structural picture). *Gerber–Norton [3] prove a bicrystal commutation at level $\ell \geq 2$ between the $\widehat{\mathfrak{sl}}_\ell$ -Kashiwara operators and their combinatorial $\mathfrak{sl}_\infty$ -crystal (also called the Heisenberg-crystal); their Corollary 5.3 predicts that in the doubly-highest-weight component the elementary Heisenberg-crystal operators send $|\emptyset, s\rangle \mapsto |\sigma[e], s\rangle$, i.e., produce partitions with all parts divisible by e . Their statements are formulated for $\ell \geq 2$; at $\ell = 1$ the Heisenberg-crystal reduces to a single-runner abacus and the closed form degenerates to $\tilde{b}_\sigma|\emptyset\rangle = |\sigma[e]\rangle$ componentwise. Theorem 4 is exactly this closed form promoted from crystal-basis elements to the class of the q -quantum vector $v_{k',e} = P_e^{k'}|\emptyset\rangle$: the q -Fock computation at $q = 0$ reproduces the Heisenberg-crystal image of the vacuum, weighted by SYT counts f^λ that record the branching of growth paths. In this sense the $\ell = 1$ case is the crystal shadow of Gerber’s Cor. 5.3; the extra f^λ multiplicity is a genuinely level-1 phenomenon coming from the ordering of the k' Heisenberg raisings that Gerber packages into a single $\tilde{b}_\sigma$.*

The bicrystal commutation, if applicable at level 1 in a suitable sense, would give $\varepsilon_i(v_{k',e}) = 0$ (Heisenberg-descendant of a Kashiwara-highest weight). What we prove is $\varepsilon_i \leq 1$, not $\varepsilon_i = 0$, and the difference is the “controlled commutation defect” predicted at the quantum level by Theorem C4 [5]: the identity $[e_i, P_e] = (q - q^{-1})[e_i, C_e^{(1)}]$ vanishes at $q \rightarrow 0$ order by order, but the $O(q^0)$ residue is non-trivial. The signature-word proof of Proposition 8 shows this residue produces at most one extra $\tilde{e}_i$ -step per residue, exactly matching $\varepsilon_i \leq 1$.

Remark 13 (Non-use of C4). The proof of Theorem 2 is independent of Theorem C4 [5]. C4 provides the quantum-level $(q - q^{-1})$ -divisibility of $e_i \cdot v_{k',e}$, and Path 2 of the PROVE.md strategy proposed an inductive proof using $R_{i,k',e} := (e_i v_{k',e}) / (q - q^{-1})$. That path is not needed: the direct combinatorial route through Theorem 4 and the signature analysis of Proposition 8 bypasses the $R_{i,k',e}$ machinery. An interesting side question, orthogonal to the current bound, is the precise relation between $R_{i,k',e} \bmod q\mathcal{L}$ and $\tilde{e}_i[v_{k',e}]$; we do not pursue it here.

7 Computational verification

All computations are pure Python + integer arithmetic; no CAS required. Code lives in `~/projects/probes/2026-` and reuses `qfock.py` (Laurent-polynomial arithmetic, standard basis, $P_e|\lambda\rangle$) and `probe4_canonical.py` (good-node combinatorics for Kashiwara $\tilde{e}_i, \tilde{f}_i$).

7.1 Phase 2 target set

Compute $\varepsilon_i(v_{k',e})$ for the target set from `state/PROVE.md`:

$$(e, k') \in \{(2, 2), (2, 3), (3, 2), (3, 3), (4, 2)\}, \quad i \in \mathbb{Z}/e\mathbb{Z}.$$

e	k'	i	$\varepsilon_i(v_{k',e})$	one-step image
2	2	0	0	—
2	2	1	1	$ (3)\rangle$
2	3	0	0	—
2	3	1	1	$ (5)\rangle + 2 (3, 2)\rangle + (2, 2, 1)\rangle$
3	2	0	0	—
3	2	1	1	$ (3, 2)\rangle$
3	2	2	1	$ (5)\rangle$
3	3	0	0	—
3	3	1	1	$2 (6, 2)\rangle$
3	3	2	1	$ (8)\rangle + 2 (5, 3)\rangle$
4	2	0	0	—
4	2	1	0	—
4	2	2	1	$ (4, 3)\rangle$
4	2	3	1	$ (7)\rangle$

All 14 triples satisfy $\varepsilon_i \leq 1$. Full log: `probes/2026-08-11-C5-gerber/epsilon_i_probe.log`.

7.2 Extended (e, k') sweep

Same test on $(e, k') \in \{(2, 4), (2, 5), (3, 4), (4, 3), (5, 2)\}$: all 15 extra triples satisfy $\varepsilon_i \leq 1$. Max $\varepsilon_i = 1$ throughout. Log: `probes/2026-08-11-C5-gerber/probe_C5_extended.log`.

7.3 Verification of the expansion (Theorem 4)

For all (e, k') with $ek' \leq 16$ (a total of 17 pairs across $e \in \{2, 3, 4, 5\}$, $k' \in \{1, 2, 3, 4, 5\}$), verify that the class $[v_{k',e}] \bmod q\mathcal{L}$ equals $\sum_{\lambda \vdash k'} f^\lambda[e\lambda]$ by explicit computation (Python integer arithmetic). All 17 pairs pass. Log: `probes/2026-08-11-C5-gerber/individual_and_expansion_check.log`.

7.4 Verification of the individual bound (Proposition 8)

For all partitions $\lambda \vdash k'$ with $k' \leq 7$ and $e \in \{2, 3, 4, 5\}$ (with $ek' \leq 20$, giving 102 distinct λ across all (e, k')), compute $\varepsilon_i(e\lambda)$ for each $i \in \mathbb{Z}/e\mathbb{Z}$. All satisfy $\varepsilon_i \in \{0, 1\}$. Distribution: 191 zero-values, 111 one-values. Log: `probes/2026-08-11-C5-gerber/individual_and_expansion_check.log`.

7.5 Signature-shape structural check

Independent verification that the raw bottom-up i -signature of $e\lambda$ always has the form $(AR)^n$ or $(AR)^nA$ (equivalently: never has an R that is not immediately preceded, in bottom-up order, by an A). Sweep: $k' \in \{0, \dots, 7\}$, $e \in \{2, 3, 4, 5\}$, all $\lambda \vdash k'$, all $i \in \mathbb{Z}/e\mathbb{Z}$. 0 failures. This directly certifies the proof structure of Proposition 8. Log: `probes/2026-08-11-C5-gerber/probe_C5_extended.log`.

8 Consequences and open questions

8.1 §7 upgrade

Combined with the previously-proved theorems T1, T2, T3, the κ -lemma (2026-08-11), and C4 (2026-08-14), Theorem 2 promotes Clio’s §7 to **six proved theorems at level 1**, with **zero remaining open conjectures** at $\ell = 1$. Higher-level lifts of C5 remain open (§8.2).

8.2 Higher-level lift

The proof of Proposition 8 is level-1-specific: it uses that at level 1 the abacus is a single runner, so “rows of μ with all parts divisible by e ” has a clean row-by-row structure. At level $\ell \geq 2$, one must work with ℓ -abaci and Gerber–Norton’s Definition 3.5 (vessels, fore/aft periods) becomes essential. We conjecture that a suitably-formulated $\varepsilon_i(v_{k',e}^{[\ell]}) \leq 1$ statement holds at higher level, provable by combining Gerber’s arrow rule (Theorem 3.14 of [3]) with a higher-level analogue of Theorem 4. This is out of scope for the current session.

8.3 Refined form (part (b) of the PROVE.md statement)

The PROVE.md refined form asked whether $R_{i,k',e} := (e_i v_{k',e}) / (q - q^{-1})$, whose existence is guaranteed by C4, has class in $\mathcal{L}/q\mathcal{L}$ agreeing with $\tilde{e}_i[v_{k',e}]$ up to a nonzero rational scalar. This is a natural next question, not needed for the string-length bound, but likely provable by combining C4’s explicit $C_e^{(1)}$ formula with the expansion of Theorem 4. We flag it as a future PROVE candidate.

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